■ LIVE ON VERCEL • HACKATHON SHOWCASE

QuickSplit IN

Zero-Knowledge Financial Settlement & Precision Cost Allocation

Built for College Project Teams, Roommates & Hackathon Participants

■■ Zero-Knowledge E2EE

■ Multi-Payer Bill Splits

■ Greedy Debt Minimization

■ 1-Click UPI & Dynamic QR

■ Live Web App: https://quickk-split.vercel.app/

PROBLEM STATEMENT

The Problem: Why Existing Split Apps Break in India

1. Multi-Payer Invoicing Chaos

College hardware purchases (Raspberry Pi, sensors, PCB printing, servers) frequently involve multiple teammates paying unequal parts on a single invoice. Existing apps strictly force only 1 payer per bill.

Inaccurate calculations & manual spreadsheet friction

2. Circular Debt Cascades

5 teammates end up sending 10 back-and-forth bank transfers instead of a simplified minimal cash flow, incurring confusion and extra UPI friction.

Unnecessary bank transfers & debt confusion

3. Cloud Database Privacy Invasions

Standard split tools store phone numbers, real names, bank UPI VPAs, and private spending logs in plain text on central cloud databases.

Leaks, scraping, and zero database privacy

4. Payment Friction & Locked QR Codes

Copying UPI IDs manually causes typos, and pre-locked QR amounts frequently fail or get rejected by Indian banking app APIs.

Broken payment deep links & payment delays

■ VALUE PROPOSITION

The Solution: QuickSplit Architecture & Pillars

■ Greedy Debt Simplifier

Min-Cash-Flow algorithm pairs net debtors with net creditors in optimal O(V) transfers. Reduces 10 circular debts to just 2 direct transactions.

Optimal O(V) Cash Flow

■ Multi-Contributor Bills

Supports bills where multiple friends contributed unevenly (e.g. Me paid ■500, Friend 1 paid ■300, Friend 2 paid ■400 on a single invoice).

Exact Split Accounting

■■ Zero-Knowledge E2EE

All member names, amounts, and UPI IDs are encrypted client-side using a salted AES cipher. The database owner sees ZERO plain-text data.

100% Private Cloud DB

■ 1-Click UPI & Dynamic QR

Direct native deep links to Google Pay, PhonePe, Paytm, BHIM, CRED, plus dynamic client-side QR codes for instant mobile camera scanning.

Instant Indian UPI Settlement

■ CRYPTOGRAPHY & ALGORITHMS

Deep Dive: Zero-Knowledge Security & Real-Time Sync

■ Zero-Knowledge Cryptographic E2EE

- **Client-Side Key Derivation:**
  Every group generates a private 12-character cryptographic key (e.g. QS-7K9A-2X4M).
- **Salted AES-Keyed Cipher:**
  Every cloud sync bundles names, bills, amounts, and UPI VPAs with an 8-character random salt and 16-bit CRC checksum.
- **Zero Cloud Visibility:**
  Firestore receives ONLY { encData: 'ENC$...' }. No plain-text names, amounts, or VPAs ever exist in the cloud database.
- **URL Fragment Transmission:**
  The decryption key travels strictly in the client-side URL hash (#group=...&key=...), completely invisible to web servers and proxies.

■ Greedy Min-Cash-Flow Algorithm

- **Net Balance Vector:**
  Calculates net credit or debt for each member across both single and multi-payer bills: Net[i] = TotalPaid - FairShare.
- **Two-Pointer Matching:**
  Concurrently matches the largest debtor with the largest creditor until all balances reach exact zero.
- **Mathematical Minimum:**
  Reduces N*(N-1)/2 complex debts down to at most N-1 clean transactions.
- **Live Bidirectional Sync:**
  Firestore onSnapshot listener updates all connected phones in real-time, backed by offline localStorage resilience.

USER EXPERIENCE

Product Walkthrough: Frictionless UX from Zero to Settled

Step 1: 30-Sec Setup

No signups, no passwords, no phone OTP. Type group name, add friends, and your workspace is live.

Zero Onboarding Barrier

Step 2: Add Expenses

Choose single or multi-contributor bill. Select split members with live per-person fair share calculations.

Ranked Category Analytics

Step 3: Direct Settle

View clear 'Who Pays Whom' banner. Tap 'Scan UPI QR' or launch Google Pay / PhonePe directly.

1-Click UPI Payment

Step 4: Share & Export

1-click formatted WhatsApp report generator + CSV / Excel export for college project reimbursement.

Audit & Submission Ready

■ COMPETITIVE ADVANTAGE

Competitive Matrix: Why QuickSplit Wins

Feature Comparison	QuickSplit IN	Splitwise	Manual WhatsApp / Excel
Account / Signup Required	■ None (Instant Web Access)	■■ Mandatory App & Account	■■ Complex Setup
Multi-Payer on a Single Bill	■ Full Uneven Multi-Pay	■■ Clumsy / Paid Pro Tier	■ Manual Math Needed
Direct Indian UPI Deep Links	■ 1-Click GPay, PhonePe, Paytm	■ No Native Indian Links	■ Manual Copy-Paste
Zero-Knowledge Cloud Encryption	■ End-to-End Encrypted (E2EE)	■ Plain-Text Server DB	■ Unencrypted
Dynamic UPI QR Generation	■ Instant Camera QR Scan	■ Static / None	■ None
Pricing & Ads	■ 100% Free & No Ads	■ Paid Subscriptions & Ads	■ Free

■ MARKET & FUTURE

Market Opportunity & Milestone Roadmap

■ High-Growth Target Audience

- **3.5M+ College Project Students:**
  Engineering & STEM teams buying IoT sensors, Raspberry Pi, robotics parts, 3D printing, and hackathon project servers.
- **15M+ Urban Roommates & Hostels:**
  Young professionals and students splitting rent, Wi-Fi bills, Swiggy /Zomato bulk food orders, and groceries.
- **Zero Customer Acquisition Cost (CAC):**
  Built-in organic viral loop: every settlement link automatically onb oards 3 to 8 new active users with zero friction.

■■ Milestone Roadmap

- **Phase 1 (Completed):**
  Multi-payer splits, Greedy settlement algorithm, UPI deep-links & dy namic QR, Zero-Knowledge Firestore E2EE sync.
- **Phase 2 (In Development):**
  Mobile Camera OCR Receipt Scanner to auto-extract line items & taxes from bill photos.
- **Phase 3 (Hackathon Scaling):**
  Web3 Decentralized Settlement & Multi-Currency support for global ha ckathon teams.
- **Phase 4 (PWA Offline):**
  Progressive Web App installation with native service worker caching.

■ QuickSplit IN

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TRY THE LIVE WEB APP NOW:

https://quickk-split.vercel.app/

Open Source • Free Forever • Built for College Hackathons & Projects